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A tunable hybrid graphene-metal metamaterial absorber for sensing in the THz regime

Mohammad Mojtaba Sadafi · Hamidreza Karami · Manouchehr Hosseini

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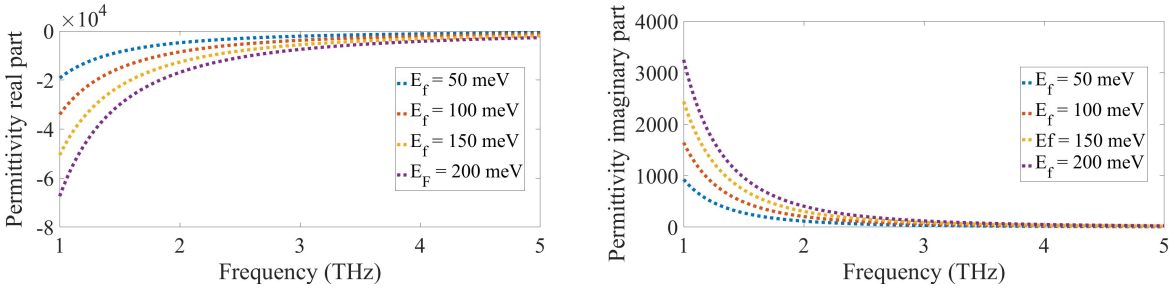
Abstract In this paper, a graphene-based metamaterial absorber is proposed and investigated numerically, in which the interaction between a split ring resonator (SRR) and graph -ene results in a high-Q absorption. To make a better understanding of the resonance mechanism, the electric and the magnetic fields, and the surface currents at the resonance frequency are investigated. In order to ease the analysis of the structure, an equivalent circuit model is introduced using the transmission line theory, and the accuracy of the proposed model is verified by the full-wave simulation. Finally, different aspects of the designed metamaterial are discussed as a potential label-free sensor for chemical and biomedical sensing. It is shown that by using this structure, a sensor with a sensitivity of 597 GHz/RIU can be achieved.

Keywords Metamaterial · Absorber · THz Regime · Graphene · Equivalent circuit model · Label-free sensor

1 Introduction

Metamaterials are artificially manufactured materials designed to obtain unique electromagnetic properties, which can not be found in conventional materials existing in nature. Metamaterials have enabled many amazing applications, such as negative refraction [1], super-resolution lenses [2,3], and cloaking [4]. One of the most beneficial features of metamaterials is the capability of electromagnetic absorption, which makes them a strong candidate for being used in EMC/EMI shielding [5]. Recently, using 2D materials has opened up the possibility of creating metamaterials in much smaller dimensions than conventional ones, having the capability of being dynamically tuned by biasing them using optical or electrical sources [6]. Graphene, as a 2D material, due to its high electron mobility, has a great thermal and electrical conductivity [7,8]. By controlling the electron mobility, one can manage a graphene sheet to have the desired conductivity, which can be achieved either by controlling the chemical doping or the gate voltage of graphene. Using graphene to create metamaterials, due to the compressed surface plasmon wavelength of graphene, has opened the way of absorption in the THz frequencies and fabricating compact devices in this frequency range, which was not possible with conventional materials [9,10,11]. THz absorbers are illustrated to have a great potential to be used in emitters [12], stealth applications[13], photodetectors [14], and photovoltaic [15]. Sensing is one of the most important applications of metamaterial absorbers [16,17,18], which is the basic concept of this paper. Sensing is a crucial procedure in order to provide information for biology and chemistry; hence, introducing a novel method with high sensitivity, robustness, and low cost has always been a matter of interest in recent research. A marker is required in conventional sensing methods, either physical or chemical, potentially altering the original properties of the material under test [19]. The response of a metamaterial strongly depends on the refractive index of the environment; this property can be used to develop label-free sensors. Defining an unknown analyte on the surface of the absorber and measuring the shift in the resonance frequency resulted from the change in the effective refractive index of the environment is the measurement procedure in metamaterial label-free sensors. Therefore, to reach a high sensing rate, the metamaterial is supposed to have a sharp response with a high quality factor. Recently, there has been numerous research on the concept of label-free sensors [20,21,22], with different aspects of limitations [23]. Graphene, as discussed earlier, due to its strong dispersion, can be used to reach a narrow-band impedance matching between the free-

Mohammad Mojtaba Sadafi · Hamidreza Karami · Manouchehr Hosseini
Department of Electrical Engineering, Bu-Ali Sina University, Hamedan, Iran
E-mail: hamidr.karami@basu.ac.ir



(a) (b)

Fig. 1: Real (a) and Imaginary (b) parts of graphene permittivity for different Fermi-levels.

space and the metamaterial structure, which is the main absorption procedure in this paper. A graphene-based structure is much smaller than those introduced in [24,25,26] and can be dynamically tuned to change the frequency range of the measurement. Furthermore, hybrid graphene-metal structures, due to their unique offered characteristics, hold noticeable research attention on different applications [27, 28,29]. To give more illustration, based on the combination of graphene and gold slits, a tunable Fano resonator with a high quality factor is introduced in [30]. Besides, a tunable graphene-metal hybrid chiral metamaterial is proposed in [31] to generate mid-infrared circular dischroism with an intensity of more than 10%. This paper introduces a hybrid graphene-metal narrowband absorber with miniaturization in size and improvement in quality factor. The small size of the absorber brings out a localized surface plasmon resonance (LSPR), guaranteeing the appropriate function as a label-free sensor. Due to its small size, the proposed sensor provides a reliable sensing rate in the presence of small amounts of analytes. Besides, a precise circuit model is derived from the transmission line theory, showing a good agreement with the full-wave simulation results. The point in using a circuit model for the structure is that one can analye and modify the introduced structure expeditiously by adding different components to the derived circuit model [32].

This paper is structured as follows: The properties of graphene are studied in Section 2, the structure of the absorber is introduced in Section 3, the circuit model is derived in Section 4, and the structure is investigated as a potential biosensor in Section 5.

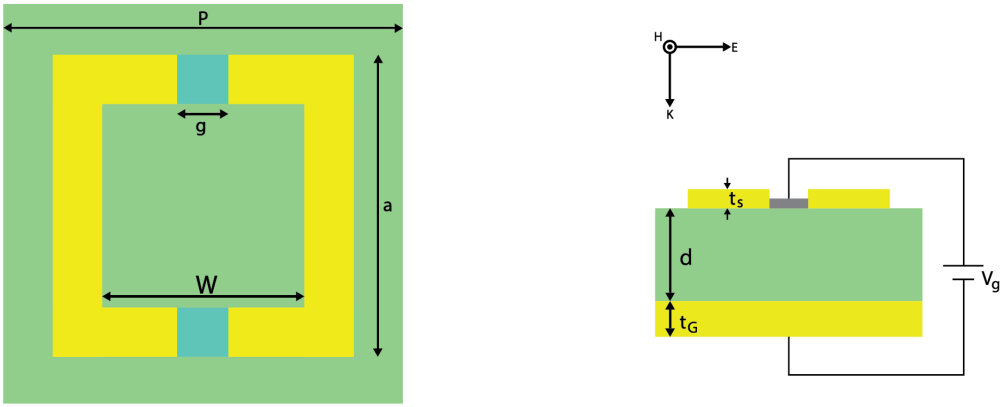
2 Graphene properties

2.1 Graphene conductivity

To better understand the absorber's performance, it is necessary to review graphene properties in the THz frequency range briefly. Using the Kubo formula, the surface conductivity of graphene can be derived as follows: [33,34]

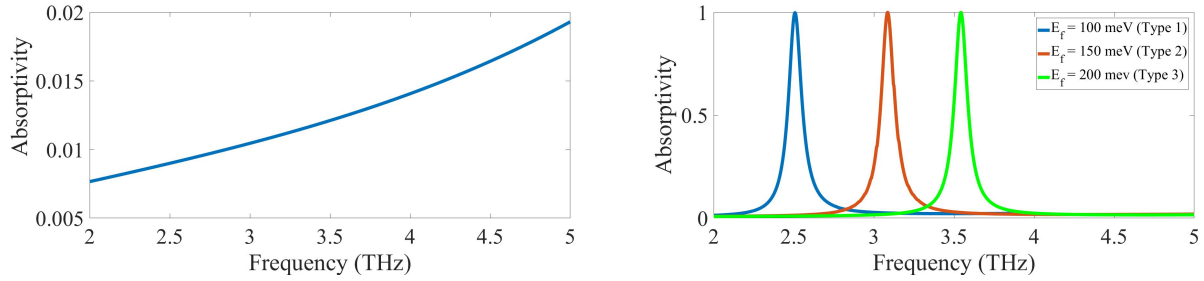
$$\sigma(\omega,\tau,\mu_c,T) =$$

$$\frac{-ie^2(\omega+i\tau^{-1})}{\pi\hbar^2}\left(\frac{1}{(\omega+i\tau^{-1})}\int_0^{\infty}\left(\frac{\partial f_d(\varepsilon)}{\partial\varepsilon}-\frac{\partial f_d(-\varepsilon)}{\partial\varepsilon}\right)\varepsilon d\varepsilon\right.$$



(a) (b)

Fig. 2: Schematic of a unit cell with related geometrical parameters for the front view (a), for the side view (b).



(a) (b)

Fig. 3: The absorption spectra obtained by the full-wave simulations for the metamaterial without graphene (a), and containing graphene with different Fermi-levels (b).

$$-\int_0^{\infty}\frac{f_d(-\varepsilon)-f_d(\varepsilon)}{(\omega+i\tau^{-1})^2-4(\frac{\varepsilon}{\bar{h}})^2}d\varepsilon\Bigg) \tag{1}$$

where e, ω, $\bar{h}$, and ε represent charge of electron, angular frequency, reduced Planck's constant, and the energy respectively. And $f_d(\varepsilon)=-\left(1+exp(\frac{\varepsilon-\mu_c}{K_BT})\right)^{-1}$ is the Fermi-Dirac distribution, $\tau=3.3\ ps$ is the electron relaxation time, and $T=300\ K$ is the temperature. The conductivity of graphene can be represented as the summation of interband and intraband conductivities as indicated below:

$$\sigma_{GR}(\omega)=\sigma_{inter}(\omega)+\sigma_{intra}(\omega) \tag{2}$$

in which each term can be derived by solving Eq(1).

$$\sigma_{inter}(\omega)=-\frac{ie^2}{4\pi\bar{h}}ln\left(\frac{2|\mu_c|-(\omega+i\tau^{-1})}{2|\mu_c|+(\omega+i\tau^{-1})}\right) \tag{3a}$$

$$\sigma_{intra}(\omega)=$$

$$-\frac{ie^2K_BT}{\pi\bar{h}^2(\omega+i\tau(-1)}\left(\frac{\mu_c}{K_BT}+2ln(exp(-\frac{\mu_c}{K_BT}+1))\right) \tag{3b}$$

where K_B is the Boltzman's constant, and μ_c reveals graphene chemical potential [35].
μ_c can be used as a means of dynamic control of graphene, and its formula is given below: [36]

$$\mu_c=E_f=\bar{h}v_f\sqrt{\frac{\pi\varepsilon_0\varepsilon_rV_g}{ed}} \tag{4}$$

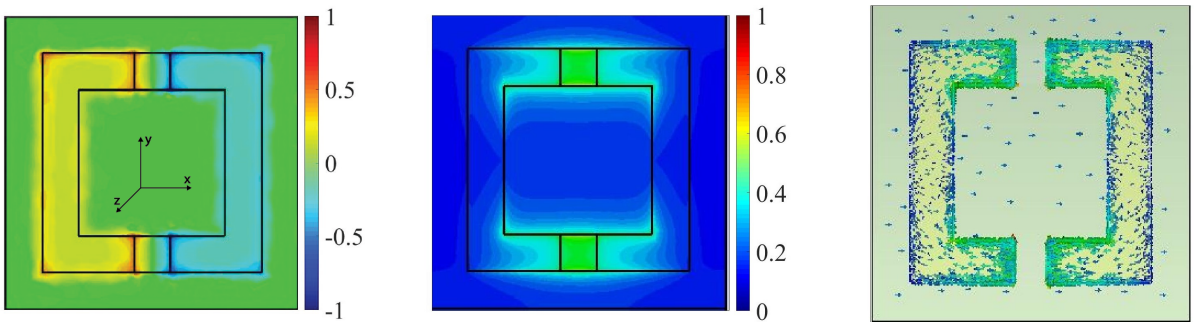
Eq(4) shows how one can dynamically control graphene properties by manipulating V_g, which will be discussed in the next section.

2.2 Graphene permittivity

A graphene sheet can be considered as a very thin slab of graphene with a thickness of $\Delta=1\ nm$ [37], and bulk conductivity of $\frac{\sigma(\omega)}{\Delta}$, so its permittivity can be calculated using the formula given below:

$$\varepsilon_{rG}=1-\frac{i\sigma(\omega)}{\omega\Delta}$$

The permittivity of graphene is shown for different Fermi-levels in Fig.1. As shown in Fig.1, in the THz frequency



(a) (b) (c)

Fig. 4: Spatial distribution at the resonance frequency for z-component of the electric field (a), the amplitude of the magnetic field (b), and surface current (c) for structure type 1 ($E_f=150\ mev$).

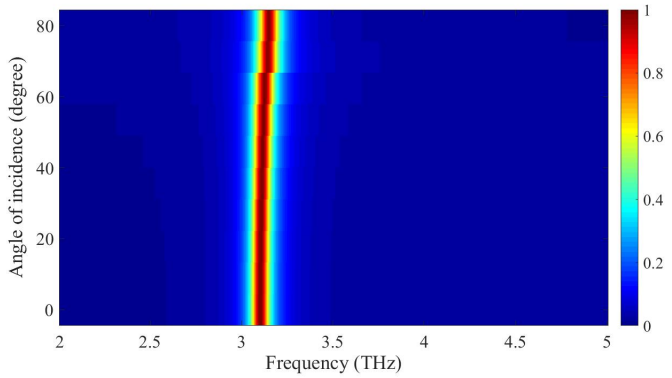


Fig. 5: Angular-resolved absorption spectra with P-polarized incidence.

range, the graphene sheet assumes negative permittivity, indicating its metallic behaviour in this range.
In this paper, CST-MWS is used as the full-wave simulator. So, the values of the permittivity can be imported to the software as the material properties.

3 Design and analysis of the absorber

3.1 Simulation and analysis

In this paper, the combination of graphene and metal [38] is used to reach a narrowband absorption peak with a high quality factor. Using graphene red makes it possible to reach absorption peaks in the THz frequency range in much smaller dimensions than metal-only resonators.
The schematic of the proposed structure is shown in Fig.2. The unit cell consists of a substrate made of SiO_2 with relative permittivity of $\varepsilon_r = 4.41$, a double-gap SRR and a slab made of gold with the conductivity of $\sigma_G = 4.561 \times 10^7\ S/m$, and two pieces of monolayer graphene filled inside the SRR gaps. Related geometrical parameters of the structure are as follows: $p = 2\ \mu m$, $a = 1.5\ \mu m$, $w = 1\ \mu m$, and $g = 0.25\ \mu m$. The thickness of the SRR is $t_s = 0.05\ \mu m$, and the thickness of the background slab, due to its role as a ground plane which is supposed to reflect all the incident wave and let no transmission, is chosen in a way to be greater than the skin depth of the gold in the THz frequency range. In our specific case, it is chosen to be $t_G = 0.5\ \mu m$. The thickness of the dielectric substrate is a variable dependent on the Fermi-level of graphene, and it should be chosen with respect to that.
In the simulation procedure of the absorber, the unit cell boundary condition is set in both the x and y directions. In addition, a Floquet port in TM00 mode (a plane wave propagating in the z-direction with the electric field in the x-direction) is used as the stimulation of the structure. Frequency domain solver (based on finite element method (based on finite element method) is also implemented in order to obtain the full-wave simulation results.)
As discussed earlier, it is possible to tune the graphene-based absorber by controlling the Fermi-level of graphene by manipulating the gate voltage. Accordingly, three different types of absorbers are proposed in this work, with three different Fermi-levels. The performance of each one is investigated, and the optimal parameters are given in Table 1. The results of the full-wave simulation are shown in Fig.3.
First of all, the response of the metamaterial without graphene is shown in Fig.3(a). Needless to say, no resonances can be achieved in the desired frequency range; it clarifies the importance of graphene to obtain absorption peaks. Furthermore, it is illustrated that by combining graphene with the structure, a tunable narrowband absorption peak with a quality factor of Q=50 can be achieved, which increasing the Fermi-level of graphene causes a clear blue shift in the absorption spectrum. To understand the underlying physics of the resonance, the effect of each element is studied separately. The ground plane acts as a reflective mirror blocking transmission and provides a broadband reflection, and the metasurface consisting of the SRR and graphene creates a plasmonic resonance, which destructive interference between them results in reduced total reflection and, therefore, enhanced absorption. Resonance in an SRR can be explained as an LC resonator. In the SRR, the split is equivalent to a capacitor, storing opposite charges at its ends. By placing graphene into the gaps, due to the high surface conductivity of monolayer graphene, charges recombine through the graphene channels and decline the capacitive effect of the split gaps [39]. This process can explain the blue shift in the spectrum that occurs with increasing the Fermi-level of graphene. To better understand the resonance, the distribu-

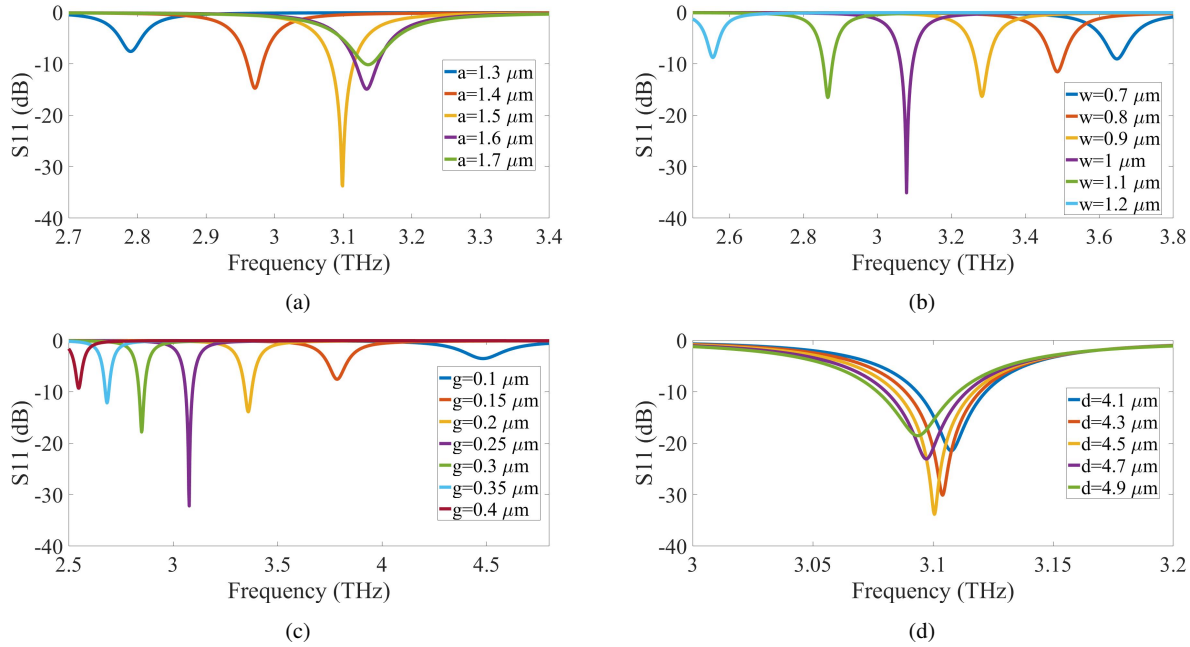


Fig. 6: Optimization of the side of the outer ring (a), the side of the inner ring (b), the length of the gaps (c), and the thickness of the substrate (d).

tions of electric and magnetic fields and surface currents at the resonant frequency are shown in Fig.4.

As can be seen in Fig.4, the resonance mode shows the characteristics of a horizontal electrical dipole [40] and an in-phase surface current distribution with a concentration close to the gaps, which leads to strong magnetic field confinement in them. Due to the strong sub-wavelength nature of the structure, and according to the surface current distribution shown in Fig.4(c), a localized surface plasmon is in charge of the resonance in the structure [41]. The performance of this plasmonic resonance, owing to the localized surface plasmon resonances, is approximately independent of the angle of incidence. The full-wave simulation results for different angles of incidence shown in Fig.5 confirm that absorption stays high for $0 < \theta < 80$.

3.2 Design procedure

As the design is supposed to be strongly subwavelength, the periodicity in both the x and y directions is chosen to be $p = 2\ \mu m$, which is about $\lambda/15$ in the THz frequency range ($1\ to\ 10\ THz$). Besides, the high operating frequency

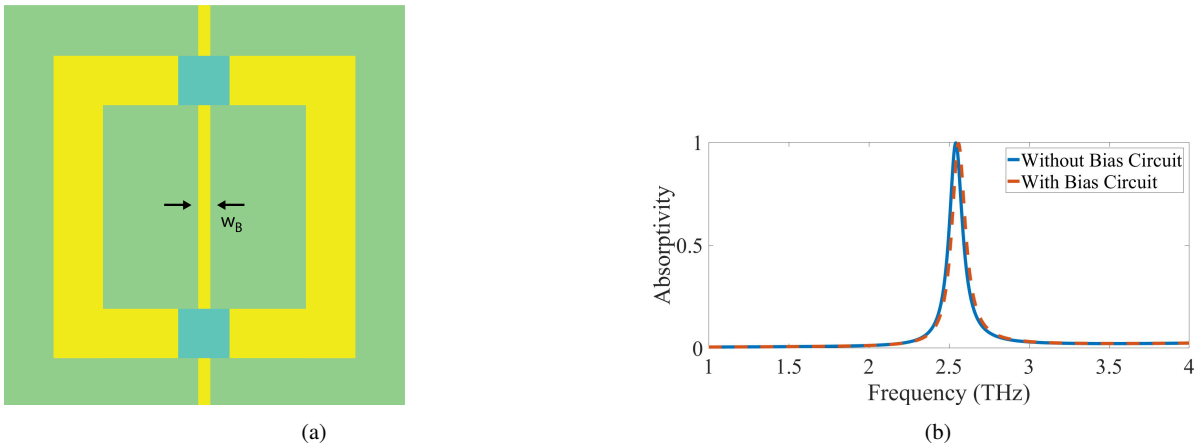


Fig. 7: Schematic of the unit cell containing the bias line (a), and the response of the absorber in the presence of the bias circuit (b).

and the presence of graphene in the structure cause conventional analytical methods of designing SRRs to lose their validity. Hence, in order to get the optimal values of the structure's parameters, different numerical optimizations are performed. The absorber is the combination of an FSS and short-circuited transmission line. The designed unit cell consists of four different parameters, where three of them (a, w, and g) are related to the frequency selective surface (FSS), and one of them (d) is the substrate's thickness. Based on Fermi-level of graphene, three types of the absorber are proposed in this paper. Due to the presence of graphene with different Fermi-levels, the parameters are optimized for the FSS, and the thickness of the substrate is used to obtain the optimal response for each type separately. The optimizations' results for the structure with $E_f = 150\ meV$ (type 2) are shown in Fig.6.

3.3 Biasing the structure

As the designed unit-cell takes advantage of two separated graphene patches, an appropriate bias circuit must be introduced to manipulate the gate voltage. In this paper, a simple approach is implemented to reach dynamic tunability. A bias line, as shown in Fig.7(a), is added to the design in order to to bias the whole structure using a simple DC voltage rail. In the subject frequency range, the skin depth of gold is about 40 nm, so the thickness of the bias line t_B is chosen to be 20 nm to minimize its impact on the performance of the absorber. Moreover, in order to tackle the manufacturing limits, the width of the bias line is $w_B = 40\ nm$. The structure's response in the presence of the bias circuit is compared with the preliminary design and shown in Fig.7(b). It can be seen that the proposed approach results in a minor error in the response of the absorber.

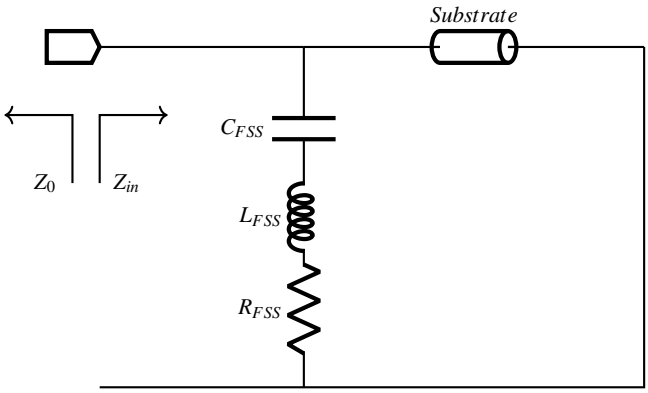


Fig. 8: The equivalent circuit model of the structure.

4 Equivalent circuit model

4.1 Deriving the model

The interaction of graphene with metal can be explained using an appropriate circuit model [42,43]. The proposed circuit model is illustrated in Fig.8. It consists of a series RLC circuit that models the frequency selective surface (FSS) and a short-circuited transmission line to model the substrate with the ground plane.
The input impedance of a short-circuited transmission line can be calculated using the formula given below:

$$Z_{ins} = jZ_s tan(k_s d) \tag{5}$$

in which $Z_s = \dfrac{Z_0}{\sqrt{\varepsilon_{SiO_2}}}$ is the intrinsic impedance of the substrate, $k_s = k_0\sqrt{\varepsilon_{SiO_2}}$ is the wavenumber inside the SiO_2, and d is the thickness of the substrate. By using the transmission line theory, the input impedance of the structure can be derived from its reflection coefficient as follows:

$$Z_{in} = Z_0 \frac{1+\Gamma}{1-\Gamma} \tag{6}$$

where Γ is the reflection coefficient obtained by using the full-wave simulation. Furthermore, the input impedance of the structure is the parallel combination of the input impedance of the substrate and the impedance of the FSS. So, the impedance of the FSS can be derived as follows:

$$Z_{FSS} = \frac{Z_{in}Z_{ins}}{Z_{ins} - Z_{in}} \tag{7}$$

It is worth noting that the impedance of the FSS should be calculated in a way to be matched with the impedance of the free-space at the resonance frequency. The numerical values for the parameters of the proposed model are given in Table 1 for different Fermi-levels of graphene (different types). The results obtained using the proposed circuit model are shown in Fig.9 for different absorber types. As is evident, they are in excellent agreement together.

Type	1	2	3
$E_f(meV)$	100	150	200
$d(\mu m)$	6.6	4.5	3.5
$R_{FSS}(\Omega)$	57.61	36.68	29.03
$L_{FSS}(pH)$	143.25	82.22	73.33
$C_{FSS}(aF)$	26.49	30.92	25.93

Table 1: Optimal lengths and circuit model parameters for different types.

4.2 Evaluation of the circuit model

A semi-numerical approach is implemented to obtain the circuit model parameters in this paper. In fact, the model takes advantage of the full-wave simulation results in the optimal length of the substrate for each type of the introduced absorber (different Fermi-levels). Once the parameters are attained, they can be used to model the structure's response in different circumstances. This semi-numerical approach would help with the integration process and dramatically decreasing of the computational cost in further analysing of the complex structures containing the proposed absorber. In order to evaluate the results obtained by the circuit model, the full-wave simulation is carried out for the structure with different substrate's thicknesses, and its results are compared to those of the circuit model and shown in Fig.10. It is clear that the proposed model is reliable in a wide range ($d = 3\ \mu m$ to $120\ \mu m$) with a negligible amount of error.

5 Analysing the absorber as a label-free sensor

5.1 The label-free sensor

Metamaterial absorbers with sharp resonances are suitable candidates for label-free sensing [44,45]. By introducing an unknown analyte on top of the metamaterial structure, due to modifying the effective refractive index of the environment, a blue shift of the resonance occurs. This frequency shift can be utilized to recognize the type of analyte. One of the most challenging problems in label-free sensing is the amount of the used analyte [23]. The proposed graphene-based absorber, due to its small dimensions, can be used as a potential label-free sensor using a much smaller amount of analyte than sensors that use only metal in their structures, which can be deployed to tackle the aforementioned problem. The effect of the thickness and refractive index of the analyte are studied in this paper. Since the refractive index of biomolecules varies from 1.4 to 1.6 in DNA and 1.6 to 2 in RNA [46], the typical refractive index is chosen to be 1.6 in this paper. The results for analytes with different refractive indices and thicknesses are shown in Fig.11.
The resonance is redshifted by 240, 336, 348, 348, and 350 GHz for analyte thickness of 0.2, 0.4, 0.6, 0.8, and 1 μm, respectively, as can be seen in Fig.11(a). As shown in Fig.11(c), the amount of redshift has an exponential dependency on the small thicknesses of the analyte. It gradually saturates for thicknesses greater than $t = 0.6\ \mu m$, which makes it suitable for sensing thin-film analytes. Moreover, it can be seen that by depositing a $0.6\ \mu m$ on the structure, a clear redshift of 7.5% occurs in the system's response. To investigate the effect of changing in the refractive index of the analyte, the simulation is carried out for an analyte with a constant thickness of $t = 0.6\ \mu m$ and different refractive indices, and the results are shown in Fig.11(b). The redshift dependency on the analyte's refractive index is illustrated in Fig.11(b). The curve shows a relatively linear dependency with a slope of 597 GHz/RIU, guaranteeing that even a small change in the refractive index of the analyte leads to a measurable redshift

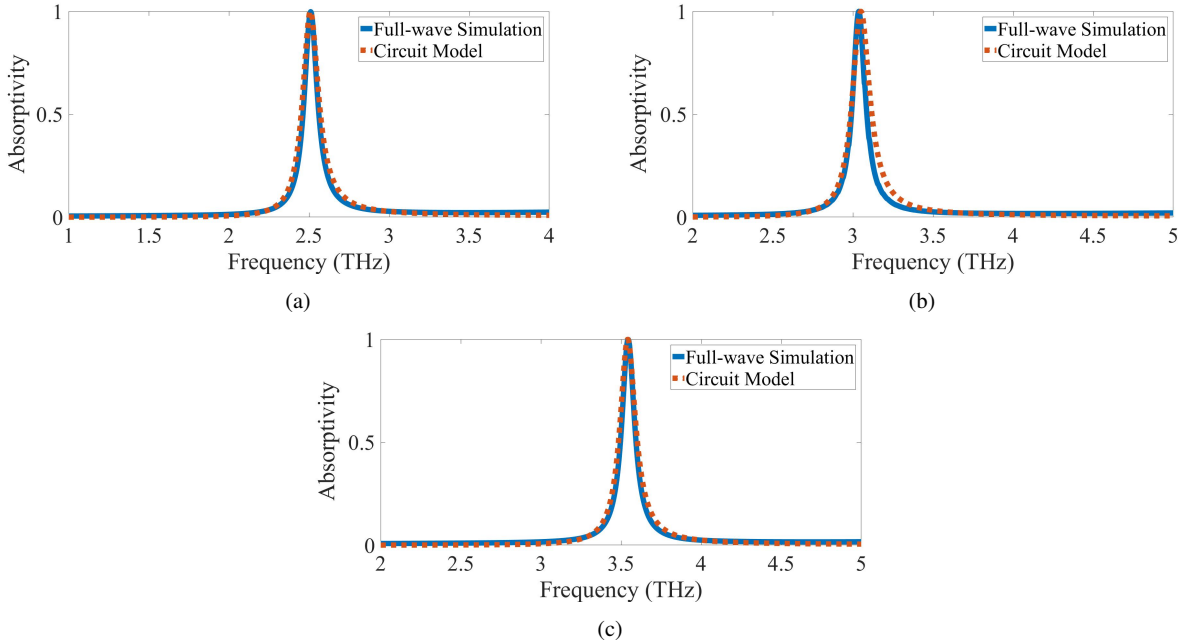


Fig. 9: The absorption spectra obtained by the full-wave simulations and circuit model for the structure of type 1 ($E_f = 100\ mev$) (a), type 2 ($E_f = 150\ mev$) (b), and type 3 ($E_f = 200\ mev$) (c).

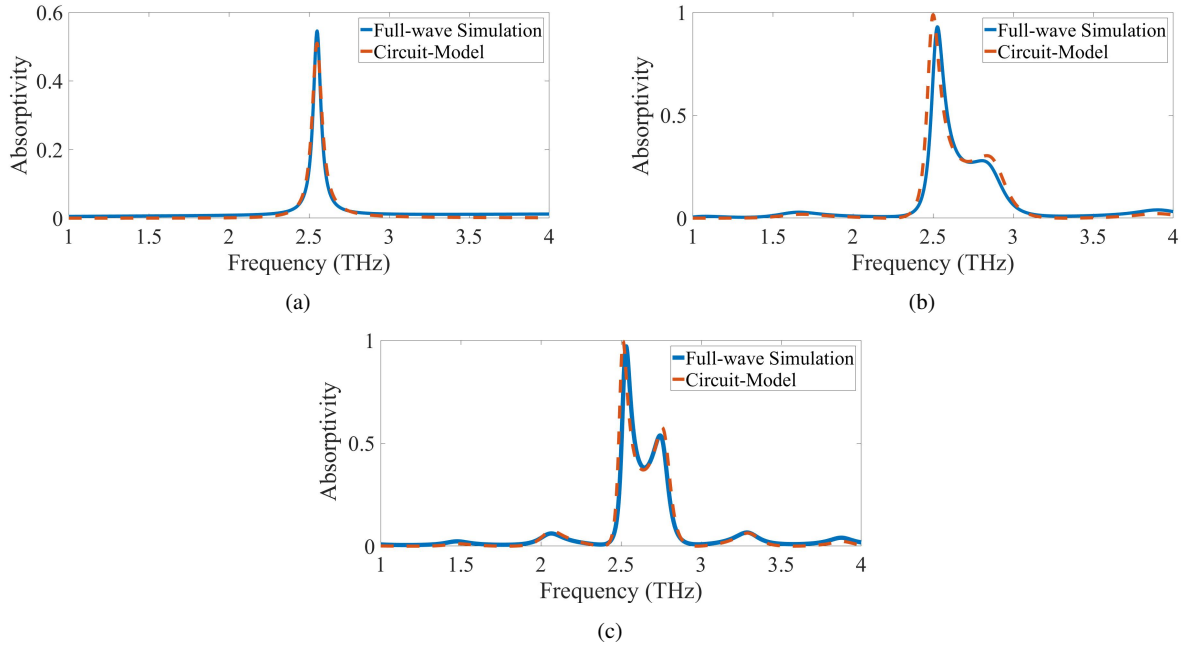


Fig. 10: Comparison between the results obtained by the full-wave simulations and the proposed circuit model for $d = 3\ \mu m$ (a), $d = 64.3\ \mu m$ (b), and $d = 120\ \mu m$ (c).

in the frequency response. Therefore, it provides an effective sensing procedure for sensing thin-film analytes based on their refractive indices.

5.2 Investigation of the previously reported sensors

To better comprehend the sensor's efficiency, a comparison between the proposed structure and the previously reported sensors is made, which its results are shown in Table.2. This comparative approach shows that due to the localized surface plasmons, the sensor has an acceptable level of sensitivity and a relatively small surface area.

In [47], a multichannel metamaterial absorber deposited on a thin, and flexible polymer is proposed, which shows a sensitivity of 54.18 GHz/RIU for the surface plasmon resonance (SPR) mode. Compared to the structure proposed in this paper, to reach the maximum sensitivity, the analyte's thickness needs to be approximately 50 μm, which is much greater than 0.6 μm in this paper.

Furthermore, in [48], a cavity metamaterial absorber is introduced, which identically as [47] benefits from an SPR. To reach the sensor's maximum sensitivity reported to be 190 GHz/RIU, an analyte with a thickness of approximately 50 μm should be used, which means a higher amount of analyte compared to this paper.

Besides, in [49], a metasurface based on free-standing asymmetric complementary split-ring resonators is proposed, which uses Fano resonance to reach a sharp response in its transmission spectra. The proposed structure represents a sensitivity of 91.7 GHz/RIU in a unit cell with dimensions of $0.3\ mm \times 0.3mm$, which is evidently greater than the dimensions of the structure proposed in this paper with a factor of 2.25×10^4.

Additionally, in [50], a dual-band complementary metamaterial composed of circular gaps is proposed in order to identify thin film refractive index and thickness. It benefits from an SPR and shows a sensitivity of 242 GHz/RIU with larger dimensions compared to this paper.

The proposed structure in [51] uses asymmetric arrangements of double split-ring resonators (DSRRs) to reach a sharp Fano resonance in its transmission spectrum. It represents a high sensitivity level of 387 GHz/RIU in dimensions of $50\ \mu m \times 50\ \mu m$, which is greater than the dimensions of the proposed structure in this paper by a factor of 6.25×10^2.

By a simple design, in [52], a metal-dielectric metamaterial absorber is proposed, taking advantage of inter-units electric resonance (IUER) to attain a high sensitivity of 2 THz/RIU in a unit cell with surface of 8100 $(\mu m)^2$ and higher working frequency range than this work.

As a new approach, based on graphene surface plasmon resonance (GSPR), promising label-free sensors are designed. To be more specific, in [53], a polarization-insensitive multiband absorber is proposed, showing a noticeable sensitivity level of 650 GHz/RIU in a relatively small unit cell. In addition, in [54], a graphene-based structure is proposed to achieve a sensor with a sensitivity of 2.06 THz/RIU in a tiny surface area, which is applicable in sensing a small

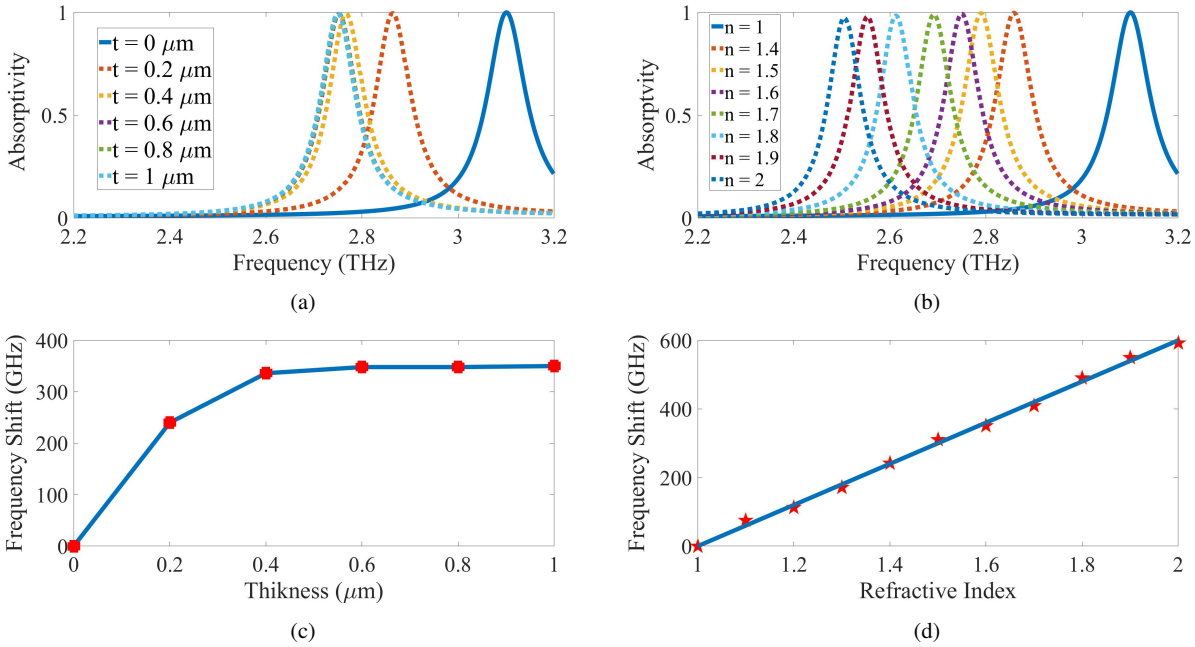


Fig. 11: Absorption spectra for analyte with refractive index of $n = 1.6$ and different thicknesses (a), and thickness of $t = 0.6\ \mu m$ and various refractive indices (b). The relation between frequency shift and thickness (c), refractive index (d).

amount of analyte in a higher frequency range than that proposed in this paper. A reliable sensor is also proposed in [55] to detect glucose and diagnose malaria. The mentioned sensor represents a very promising sensitivity of 22.72 THz/RIU in a unit cell, about two times bigger than this paper and a higher working frequency range.

6 Conclusion

In this paper, a tunable narrow-band absorber has been introduced. Besides, it has been shown numerically that the perfect absorption can be achieved in the THz frequency range in tiny dimensions using graphene. An accurate circuit model has been proposed based on the LC behaviour of the resonance, and it has been demonstrated that the results of the full-wave simulation are in perfect agreement with the results obtained by the proposed circuit model. Furthermore, the proposed absorber has been investigated as a label-free biosensor, and it has been shown that a high sensitivity of 597 GHz/RIU can be achieved, which the proposed structure a potential device for sensing very small amounts of analytes. It has also been shown that in comparison with the previously reported sensors, the proposed sensor shows a higher level of sensitivity in much smaller dimensions.

Acknowledgements The authors would like to to give their concrete acknowledgement to Erfan Seif for his helpful graphical designs.

Conflict of interest

The authors declare that they have no conflict of interest.

Mode	Surface Area	Sensitivity	Q	FoM	Reference
SPR	$9 \times 10^4\ (\mu m)^2$	54.18 GHz/RIU	N.A	N.A	[47]
SPR	$2.25 \times 10^4\ (\mu m)^2$	190 GHz/RIU	11.6	2.3	[48]
Fano	$9 \times 10^4\ (\mu m)^2$	91.7 GHz/RIU	29	6.1	[49]
SPR	$16 \times 10^4\ (\mu m)^2$	242 GHz/RIU	40.2	4.82	[50]
Fano	$25 \times 10^2\ (\mu m)^2$	387 GHz/RIU	N.A	N.A	[51]
IUER	$8100\ (\mu m)^2$	2 THz/RIU	59.8	40	[52]
GSPR	$196 \times 10^2\ (\mu m)^2$	650 GHz/RIU	50	5.3	[53]
GSPR	$6.25\ (\mu m)^2$	2.06 THz/RIU	12.1	3.9	[54]
GSPR	$8.35\ (\mu m)^2$	22.72 THz/RIU	14.57	34.96	[55]
LSPR	$4\ (\mu m)^2$	597 GHz/RIU	49.2	9.48	This Paper

Table 2: Comparison between the proposed sensor and the previously reported sensors.

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